



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination
Academic Year: 2025-26 (semester I)

Class: Third Year B. Tech.

Course: Fuel Cell Engineering

Course Code: PEC-CH309

Date: 18-08-2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. *All questions are compulsory.*
- ii. *Assume suitable data whenever necessary and state it clearly.*
- iii. *Figures to right indicate full marks.*
- iv. *The symbols carry usual meaning.*

Q. N.			Marks	CO	BT
Q.1	a	What is the fundamental electrochemical working principle of a fuel cell?	3	CO1	BT1
	b	What are the advantages and limitations of fuel cells?	3	CO1	BT1
	c	What is a Proton Exchange Membrane Fuel Cell (PEMFC) and how does it work?	4	CO1	BT2
Q.2	a	What are the different types of fuel cells and their applications?	3	CO2	BT3
	b	Explain the Solid Oxide Fuel Cell (SOFC) and its working principles?	3	CO2	BT3
	c	Describe the Alkaline Fuel Cell (AFC) and its working principles?	4	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Introduction to fuel cell
CO 2	Fuel Cell Types and Components
CO 3	Differentiate between various types of fuel cells and their respective advantage, limitations, and applications
CO 4	Design and analyse fuel cell components and operation conditions for enhanced performance.
CO 5	Assess fuel processing methods for hydrogen production and alternative fuel utilization in fuel cells.
CO 6	Critically evaluate the challenges, economic feasibility, and future prospects of fuel cell technology in a sustainable energy system



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
ISC -I- Examination (Semester-I) 2025-26

Department : Chemical Engineering

Class : T.Y. Chemical

Chemical Reaction Engineering -I

Sub. Code: PCC-CH305

Date: 20/08/2025

Time: 15:00 -15:45

Max. Marks :20

Note: i. Attempt all questions

ii. Assume suitable data if necessary iii. Use of non-programmable calculator is permitted.

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Q.N	Questions	Marks	CO	BT	
Q1	a	What do you understand by Order of Reaction? Explain with suitable example	03	CH302:1	1
	b	Which variables are affecting on the reaction rate? Explain in brief.	03	CH302:2	2
	c	Milk is pasteurized if it is heated to 64°C , for 35 min, but if it is heated to 74°C it only needs 20 s for the same result. Find the activation energy of this sterilization process	04	CH302:1	5
OR					
	c	What do you understand by Chain reaction and chain reaction? Explain with suitable example.	04	CH302:1	1&2
Q.2	a	What are the methods of analysis of kinetic data? Derive the integral form for Second order reaction in case of concentration and conversion	05	CH302:1	3
OR					
	a	In a homogeneous isothermal liquid polymerization, 20% of the monomer disappears in 34 minutes for initial monomer concentration of 0.04 and also for 0.8 mol/liter. What rate equation represents the disappearance of the monomer?	05	CH302:2	5
	b	What do you understand by constant volume batch reactor? Give the types of ideal reactor.	03	CH302:	3
	c	What do you understand by activation energy? give its significance	02	CH302:1	1

Course Outcomes (CO)

CO-1	Understand fundamental concepts of reaction engineering, type of reactions, and reaction kinetics.
CO-2	Enhance their knowledge of reactors, the working of reactors, and different types of arrangements of reactors.
CO-3	Improve the students ability to decide the type of reactor and their arrangement or achieving the desired conversion while keeping the economy in point of view.
CO-4	Analyze series and parallel reactions, compute yield/selectivity, and design reactors accordingly.
CO-5	Solve energy balance equations and evaluate the impact of temperature variation on reactor performance
CO-6	Solve numerical and conceptual problems in reaction engineering, preparing for industrial applications and advanced studies.

- PCC-CH302:1 To get the knowledge about the basic concepts of reaction kinetics, rate and order of reactions.
- PCC-CH302:2 Able to calculate material, energy and behavior of the reaction balance for reactors.

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Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY

Course: Material Science and
Engineering (Elective)

Course Code: PEC-CH305

Date: 18.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. Figures to right indicate full marks.

Q.N.		Question	Marks	CO	BT
Q.1	a	Define material science? Explain classification of material.	05	CO1	BT1
Q.2	a	What is Cast iron? What is composition of cast iron? Explain manufacture of cost iron with net label diagram.	05	CO1	BT2
Q.3	a	What is iron ore? Explain type of iron ore, properties, formulation & uses.	05	CO1	BT1
Q.4	a	Write a note on. I. Grey cast iron II. White cast iron.	05	CO1	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify and differentiate between various ferrous metals based on their properties and uses.
CO 2	Explain the significance and applications of non-ferrous metals in engineering and industry.
CO 3	Interpret phase diagrams and understand the strengthening techniques of metal alloys.
CO 4	Describe the classification, composition, and industrial significance of glass.
CO 5	Analyze polymer properties, polymerization methods, and their applications in different fields.
CO6	Evaluate the properties and applications of composite and nanomaterials in modern technology

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded ISE I, Sem-V, AY 2025-26 Class: Third Year B. Tech. (Chemical Engineering)	
Date: 19/08/2025 Time: 15:00 to 15:45	Subject: Mass Transfer-II (PCC-CH303)	Max. Marks: 20

- Note:
- Attempt all questions.
 - Assume suitable data if necessary.
 - This paper contains one page.

Sr. No	Question	Marks	CO	BT														
Q.1	Define the following terms 1. Absorption 2. Height of transfer unit (HTU) and Number of transfer unit (NTU) 3. Height equivalent to theoretical plate (HETP)	3	PCC-CH303.1&2	BT1														
Q.2	What are the general criteria for the selection of a solvent for absorption. Explain them in detail.	5	PCC-CH303.1	BT2														
Q.3	H ₂ S is to be absorb from an air mixture counter-currently in a plate column with fresh water. H ₂ S concentration is reduced from 10 mole % to 1 mole % and the inlet water and gas rates are 500 and 400 kg/m ² s respectively. Calculate the number of actual plates required for the absorber. Equilibrium data for the system at 20 °C and 1 atmosphere is: <table><tr><td>X</td><td>0.0050</td><td>0.0164</td><td>0.0252</td><td>0.0349</td><td>0.0455</td><td>0.0722</td></tr><tr><td>Y</td><td>0.0054</td><td>0.0210</td><td>0.0320</td><td>0.0420</td><td>0.0533</td><td>0.0800</td></tr></table> Where, X is kmol H ₂ S/kmol H ₂ O and Y is kmol H ₂ S/kmol Air	X	0.0050	0.0164	0.0252	0.0349	0.0455	0.0722	Y	0.0054	0.0210	0.0320	0.0420	0.0533	0.0800	6	PCC-CH303.2	BT5
X	0.0050	0.0164	0.0252	0.0349	0.0455	0.0722												
Y	0.0054	0.0210	0.0320	0.0420	0.0533	0.0800												
Q.4	Derive the design equation for the minimum liquid flow rate in a gas absorption process. Illustrate the derivation with a diagram and explain its significance	6	PCC-CH303.2	BT4														

Course Outcomes (CO)

- PCC-CH303.1 Analyze gas-liquid equilibrium of absorption and stripping towers.
PCC-CH303.2 Design absorption and stripping towers using mass transfer correlations and stagewise calculations.

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Department of Chemical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Third Year

Course: Plant Utility

Course Code: PEC-CH303

Date: 18.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. Solve any Two questions.
- ii. Figures to right indicate full marks.

Q.N.			Marks	CO	BT
Q.1	a	What is process plant utility? Write down its importance.	05	CO1	BT1
	b	Write the sources of water and its characteristics.	05	CO1	BT1
Q.2	a	Enlist the methods of water treatment and explain the reverse osmosis process in details with neat labelled diagram.	10	CO2	BT2
Q.3	a	What do you understand by the term priming and foaming, with its advantages and disadvantages.	06	CO2	BT2
	b	Write the prevention of priming and foaming	04	CO1	BT1

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

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Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand Industrial Water Treatment – Explain water properties, hardness, boiler feed issues, and water softening methods.
CO 2	Explain Steam & Non-Steam Heating Systems – Describe steam properties, boilers, mountings, accessories, and Indian Boiler Act.
CO 3	Analyse Refrigeration Systems – Understand refrigeration concepts, cycles, refrigerants, and performance evaluation
CO 4	Apply Psychrometry Principles – Interpret psychrometric properties, processes, and cooling tower operations
CO 5	Evaluate Insulation Techniques – Explain insulation materials, their importance, and applications in process industries
CO6	Understand Plant Maintenance Strategies – Learn types of maintenance, responsibilities, and online equipment maintenance.



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Department of Chemical Engineering
In-Semester Examination I
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Class: Third Year

Course: Plant Utility

Course Code: PEC-CH303

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CO 5	Evaluate Insulation Techniques – Explain insulation materials, their importance, and applications in process industries
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